SCI-PTC

Correlated-Mode Cleaning and Detector Coefficients

Engineering Conformance Specification

Scientific contract v0.1 – document revision r0.5 candidate

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Status. Scientific-owner review candidate; not frozen. The frozen predecessor remains v0.1/r0.4. This specification defines candidate conformance obligations but reports no implementation assessment, validation result, or production authorization.

Normative rule. The six shared modules imported below are the sole normative science. This view adds organization and evidence guidance only; it does not introduce an independent equation, estimator, threshold, or scientific interpretation.

Contents

1 Conformance scope and evidence posture

Conformance means that a candidate implementation, its interfaces, and its published products satisfy every applicable shared requirement for the exact realized product role. A source trace or readable file alone is not sufficient. Evidence must bind numerical behavior to immutable identity, declared supports, realized state, response, uncertainty status, and output role.

The present document provides no implementation mapping and no pass/fail disposition. A later authorized conformance effort shall classify each requirement as conformant, nonconformant, partial, unavailable, or not applicable with exact evidence and scope. Unknown is never a pass.

Evidence layer	Required interpretation
Algebraic contract	The shared definitions, equations, assumptions, requirements, and predictions are the authority to be satisfied.
Implementation conformity	Requires source/interface/product evidence for every applicable requirement; unassessed here.
Representation fidelity	Requires proof that serialized state, support, identity, and response represent the realized numerical operation; not performed here.
Observational performance	Requires approved data and thresholds for residual contamination, sky transfer, stability, covariance, and source recovery; not performed here.
Production readiness	Requires separate operational evidence and authorization; not assessed here.

2 Required interface families

An implementation claiming SCI-PTC v0.1 conformance must expose or publish the following scientific interface families. Exact representation is engineering- owned, but every scientific field below must be recoverable without guessing.

Interface family	Required recoverable content
Input identity	Immutable CAL parent; complete RTC parent; observation, scan, segment, sample-time, detector occurrence/UID, array, group, stage, product role, pass, recurrence, and parent identities; exact axes and storage mapping; typed source/residual or surrogate parent where used.
Signal contract	Top-of-atmosphere, point-source-equivalent mJy per fixed nominal beam; factor/atmosphere lineage; validity, uncertainty, nuisance-correlation, and complete upstream-response status.
Support policy	Complete accumulated producer facts and causes without erasure; for each distinct PTC-owned use, the complete required-restriction set, T/F/U/C knowledge, exceptions, total decision rule, missing/conflict behavior, scope, owner, use, and policy/version identity; fit-excluded/apply-allowed state and exact group-local inputs required to define it. Downstream facts remain owned by their named-use owner.
Estimator request	Family, fit objective/loss/metric, missing-data model, configured network- or array-level grouping, group-local masks/support, ordinary detector-wise support-normalized arithmetic-mean centering on finite basis-fit support, identity scaling, strictly positive rank, gauge, source-protection state, zero-refinement state, response, covariance, simulation, and outputs. Future automatic-selection families additionally require candidates, ordering, ties, and typed predicates.
Learned evidence	For the ordinary route, the one requested positive-rank candidate per group, fitted or failed state, feasibility cause, and diagnostics; for a future automatic family, every candidate, predicate, threshold, uncertainty, ordering/tie result, and disposition.

Interface family	Required recoverable content
Resolved model	Exact admitted identities and configured groups; fitted basis/loadings; scaled correlated and total removed signals; detector-right acting space; mask-aware coefficient solve; strictly positive rank; degeneracy/gauge; learned centering location and nonrestoration; identity scaling; three null/loss roles; exact frozen application map and derivative/linear part; failure and stop state.
Coefficient products	Separate fitted loadings, coordinate parameters, diagnostics, and detector- or detector-time-indexed analysis/gridding families with units, factor domains, normalization operator/domain, support, lifecycle, numerical use, consumers, and prohibited interpretations.
Response products	PTC-local operator, complete admitted upstream response ending on the CAL grid, complete upstream-to-PTC chain response, realized propagated companions with exact parent domain, full-PTC-procedure signal-response family and typed state differences or explicit unavailability, and optional map-center functional with exact response role and reference-MAP parent.
Uncertainty products	Exact status and conditioning for formal covariance, marginal variance, empirical scatter, calibration/systematic, response, selection, cross-coordinate, and cross-observation terms.
Output/provenance	Transformed TOD role; immutable requested, effective-policy, observation-resolved, learned, resolved, applied, and published state; group identity; product-realization and role-availability axes after PTC is entered; RTC-terminal route publication and terminal state when PTC is disabled; required siblings, completeness, failure scope, random/surrogate state, and typed relations to downstream consumers.

3 Fail-closed implementation behavior

Failure is scoped to the affected product or claim unless a required product dependency makes the entire requested output incomplete. Implementations shall represent missing, non-finite, invalid, rejected, disabled, not requested, unavailable, and numeric zero as distinct states. Product realization, response availability, covariance availability, and evidence disposition are independent axes. Implementations shall not infer missing units, identities, supports, thresholds, rank rules, response terms, bias claims, or covariance.

For every PTC-owned named use, a support decision is conformant only when it evaluates the owner's complete required-restriction set under the total T/F/U/C rule after request, binding, and applicability checks. Any decisive false yields `ineligible`; all true yields `eligible`; every other applicable requested state yields `decision_unavailable`. The trace retains every cause, unknown, conflict, exception, scope, owner, use, and policy/version. Shared VAL machinery may evaluate this supplied policy but shall not invent it. Permission for one use cannot substitute for another.

A typed fallback is conformant only when the request authorizes it and the realized state records its cause, stage, changed scientific operation, response consequence, covariance consequence, and affected availability. Required-output failure propagates to the calling boundary. Best-effort diagnostic failure is inert only when that diagnostic is neither required nor used by policy.

For the ordinary route, k must be an explicitly requested integer greater than or equal to one and realizable in every required group. Rank zero and an unrealizable positive rank fail without clipping, substitution, centering-only output, disabled-route conversion, RTC export, or map. The route performs one immutable-parent fit per group and zero support-changing refinements.

An explicit PTC-disabled request instead requires complete RTC publication and successful termination after RTC without entering CAL, PTC, or MAP. Failure of required RTC publication is a route failure. Library code returns the typed terminal result rather than implementing process-level exit.

The base-v0.1 conditioned-r branch is diagnostic-only. A conformance trace must show that it cannot change calibrated-x fit membership, requested or resolved rank, subtraction, output, or coefficients. Raw-r processing, r-derived subtraction from x, joint x/r PCA, raw Beammap PTC, cross-array fitting, and polarimetry are outside this contract.

4 Evidence checklist for a later assessment

For each claimed product role, a later assessment should obtain:

1. the immutable requested, effective-policy, observation-resolved, learned/fitted, resolved-selected, applied/realized, and published states and their parent links;
2. a field-level source/interface trace for every shared requirement;
3. minimal deterministic probes for the applicable shared predictions;
4. exact comparisons of fitted support, candidate evidence, removed subspace, frozen application map, classification, response-domain chain, covariance status, expectation/bias status, and coefficient semantics;
5. proof that ordinary PCA/SVD uses the configured network- or array-level group, group-local support and masks, detector-right mask-aware coefficient recomputation, and strictly positive requested rank; that the learned λ_g is subtracted from science data and not restored; that fixed-state companions hold λ_g fixed without subtracting it from the perturbation; that full-procedure response, when separately requested, re-estimates and again discards λ, that fixed-state companions cannot affect learning, that the complete upstream response ending on the CAL grid is applied exactly once, and that full-procedure response restarts from the immutable CAL parent while retaining typed state changes;
6. proof that every fit-excluded/apply-allowed occurrence has the exact basis-coefficient or recomputation input, neighboring support, detector binding, metric, coordinate, and boundary state required by the resolved operator, and otherwise becomes unavailable without inferred interpolation, zero filling, reconstruction, or extension;
7. failure probes for missing external-parent identity, mismatched surrogate membership, zero/empty support, non-finite values, $k = 0$, unrealizable positive rank, forbidden rank clipping, cross-group support leakage, disabled-route RTC publication failure, unavailable response, independent realization/availability states, and incomplete publication; and
8. an evidence-layer disposition that does not promote conformance into observational performance or production authorization.

The shared prediction register is deliberately implementation-independent. A later probe may falsify a conformance claim, but passing a small synthetic case does not by itself validate observational performance. The science-team validation table groups the intended studies and observables; it supplies no threshold, result, or evidence-layer promotion.

5 Normative notation

Symbol	Meaning
o, s, g	Immutable observation, scan, and coherent-segment identities.
t, d	Sample position on the declared time grid and immutable detector occurrence/UID.
a, n, ℓ	Array, readout network/electronics group, and optional local or focal-plane group.
q	Immutable product-stage identity; it is never a diagnostic statistic.
$Y \equiv Y^{\mathrm{CAL}}$	Admitted sample-by-detector calibrated x matrix, in top-of-atmosphere point-source-equivalent mJy per fixed nominal beam.
S	Astronomical contribution on the admitted detector-time grid.
$M_\star = [m_{\star,tk}]$	Latent shared temporal-mode or supplied-template representation admitted by the signal model.
$A_\star = [a_{\star,dk}]$	Latent detector-loading representation for the admitted correlated component.
$U_\star = M_\star A_\star^\intercal$	Correlated-mode or supplied-template component admitted as the estimand; it is not yet a fitted estimate.
$\widehat{\widehat{U}}_{\mathrm{corr}} = \widehat{M}\widehat{A}^\intercal$	Realized fitted estimate of the correlated component in the declared scaled fitting coordinate.
$\widehat{U}_{\mathrm{corr}} = D\widehat{\widehat{U}}_{\mathrm{corr}}$	Realized fitted correlated component restored to the admitted CAL physical unit; correlation does not identify its physical origin.
$U_{\mathrm{total},\Theta}(Y') = Y' - \mathcal{A}_\Theta(Y')$	Total removed signal defined by the input-output difference for the exact application operator; for the nonrestoring linear specialization it contains both the discarded λ and the correlated-subspace component.
$Z \equiv Y^{\mathrm{PTC}}$	Transformed calibrated detector timestream.
N	Remaining detector-noise term; it is not assumed independent or stationary.
$g, \mathcal{G}$	One configured PCA group and the selected group set: one full array for array-level PCA or the set of readout networks in that array for network-level PCA.
C_λ, D, λ	Nonrestoring centering map $C_\lambda(Y) = Y - \lambda$, invertible internal scaling operator, and declared learned additive-location state.
$\mathcal{T}^{\mathrm{ctr}}_{g,d}, w^{\mathrm{ctr}}_{gtd}$	Exact finite basis-fit-admitted time occurrences used to center detector d in group g, and their ordinary-route binary influence weights; no inverse-noise or other numerical weighting is implied.
$\widehat{\mathcal{U}}$	Realized removed subspace, represented independently of basis gauge.
$\mathcal{N}_{\mathrm{PTC}}$	Declared PTC null space, including additive-reference losses.
$\mathcal{S}^j$	Support for use j: basis fit, loading fit, application, output, coefficient/QC, response, empirical, simulation, or downstream use.
$\Phi, \mathcal{I}, \mathcal{C}$	Typed direct causes, transitive causal-influence record, and the accumulated cause/fact set or graph retained without erasure.
$U, \mathcal{R}_U, K(r)$	Exact named scientific use, its complete owner-controlled required-restriction set, and the T/F/U/C knowledge state of restriction r.
$\mathcal{M}_{\mathrm{cand}}, c, c_\star$	Finite candidate-model-specification set, one candidate specification, and the selected specification before resolution into Θ; $\mathcal{C}$ is never reused for this role.
$W^{\mathrm{fit}}_g, W^{\mathrm{app}}_{g,t}$	Exact group-local fit-influence and time-local application-influence masks or weights; zero influence is not a numerical zero observation.
$k_{\mathrm{req},g}$	Explicitly requested strictly positive PCA rank for group g; zero is invalid and is not PTC-disabled.
R, E, O	Requested, effective-policy, and observation-resolved input states.
$\mathfrak{L}$	Complete learned/fitted evidence for every evaluated candidate, before final selection.
Θ_g	Immutable group-local resolved selected model, including exact grouping level, support, masks, metric, rank, subspace, degeneracy, generalized inverse/tolerance, application map, and selection or feasibility evidence.
Λ, Π	Applied/realized state and published product state.

Symbol	Meaning
$\mathcal{A}_{\Theta_g}$	Exact frozen-state numerical application map acting on one admitted group-local detector-time input.
$J_{\Theta_g}[Y_g]$	Fréchet derivative $D\mathcal{A}_{\Theta_g}[Y_g]$, or the declared finite-dimensional linear part when $\mathcal{A}_{\Theta_g}$ is affine.
$\mathcal{L}^t_\Theta, \mathcal{L}^d_\Theta, \mathcal{L}^v_\Theta$	Frozen projection operators in temporal, detector, or explicitly vectorized detector-time space.
$\mathcal{P} = (\mathcal{P}_Z, \mathcal{P}_\Lambda)$	Full data-dependent PTC procedure and its transformed-signal and typed-state projections.
$K_{\mathrm{up}\to\mathrm{CAL}}$	Complete admitted upstream response carried by the CAL product from a declared source-domain perturbation to the CAL detector-time grid, including every applicable admitted source-to-detector, beam-convention, coordinate/scan, RTC, and CAL operation; otherwise an explicit unavailable state.
τ_{src}	Exact named source-domain perturbation/template.
$k^{\mathrm{CAL}}_{\mathrm{src}}$	Realized source companion already on the admitted CAL detector-time grid; when source-domain propagation is requested, $k^{\mathrm{CAL}}_{\mathrm{src}} = K_{\mathrm{up}\to\mathrm{CAL}}\tau_{\mathrm{src}}$.
$k^{\mathrm{PTC}\mid\Theta}_{\mathrm{src}}$	Fixed-state propagated detector-time companion $J_\Theta[Y]k^{\mathrm{CAL}}_{\mathrm{src}}$.
$K_{\mathrm{PTC}\mid\Theta}$	Complete fixed-state chain response $J_\Theta[Y] \circ K_{\mathrm{up}\to\mathrm{CAL}}$.
$K^{\mathrm{proc}}_{\mathrm{PTC}}$	Full-PTC-procedure finite-perturbation signal-response family from the immutable CAL parent.
Δ_{state}	Typed comparison of two discrete or mixed realized states; it is not numerical subtraction.
A_{ref}	Exact named reference MAP functional, imported only for an optional diagnostic.
ρ_{center}	Optional estimated map-center point-source response diagnostic.
C_Y, C_Z	Declared input and transformed-signal covariance, with exact status and conditioning.
η_i	Declared diagnostic statistic for coefficient index i, where $i = d$ or $i = (t, d)$ according to the coefficient family.
γ_i	A downstream analysis/gridding coefficient with declared detector or detector-time index family; never a fitted loading by implication.
Avail_r	Typed availability for requested role r: `computed_published`, `not_computed_or_not_requested_for_this_product`, `invalid`, or `unavailable`.
Exist_p	Product realization state for product p: `realized`, `disabled`, `not_requested`, `rejected`, or `unavailable`.

Superscripts and subscripts identify scientific role and conditioning, not a storage layout. Dense row or column positions are local conveniences and are never stable external identities. Operator products are written only when their domains and codomains are compatible.

6 Normative definitions

`SCI-PTC-DEF-001` – **Admitted CAL parent.** The immutable SCI-CAL detector-time product accepted by all required unit, identity, validity, lineage, uncertainty, and response gates. It is the numerical parent of every base-v0.1 fit and support-changing refit.

`SCI-PTC-DEF-002` – **Correlated-mode estimand.** A latent declared detector-time component $U_\star$ represented by a shared temporal basis or supplied template and detector loadings. PTC estimates $\widehat{\widetilde{U}}_{\mathrm{corr}} = \widehat{M}\widehat{A}^{\mathsf{T}}$ in the scaled fitting coordinate and $\widehat{U}_{\mathrm{corr}} = D\widehat{\widetilde{U}}_{\mathrm{corr}}$ in the admitted CAL physical unit, plus the realized removed subspace $\widehat{\mathcal{U}}$. The operator-defined total removed signal is distinct, and statistical correlation does not identify the physical origin of either removed quantity.

`SCI-PTC-DEF-003` – **Transformed detector signal.** The operator-primary result $Z = \mathcal{A}_\Theta(Y^{\mathrm{CAL}})$ after discarding the learned additive detector reference, applying the realized correlated-mode operation, and restoring only invertible internal scaling. The total removed signal is defined as $U_{\mathrm{total},\Theta}(Y^{\mathrm{CAL}}) = Y^{\mathrm{CAL}} - Z$ rather than identified with the fitted correlated component alone. The discarded reference carries no scientifically meaningful optical DC information. The result is not an independent sky estimate.

`SCI-PTC-DEF-004` – **Removed subspace.** The complete realized subspace or template span whose fitted contribution is rejected, including its configured network- or array-level group, support, masks, metric, strictly positive rank, order, degeneracy, generalized-inverse state, and gauge-equivalence class.

`SCI-PTC-DEF-005` – **Null and loss roles.** Three separately published roles: the fixed-state linear null $\ker J_\Theta$; modes made invariant or unidentifiable when centering is re-estimated by a declared full procedure; and the affine additive-reference loss caused by discarding λ. A sky component overlapping the removed subspace may appear in one or more roles, but the roles are not interchangeable.

`SCI-PTC-DEF-006` – **Gauge.** A sign, scale, ordering, or rotation freedom in a basis/loading representation that leaves the fitted correlated component and removed subspace unchanged.

`SCI-PTC-DEF-007` – **Centering state.** For the ordinary route, the detector-wise time-axis additive location $\lambda_{g,d}$ is the support-normalized arithmetic mean of finite basis-fit-admitted occurrences for detector d within one exact immutable PTC segment and configured group. Its centering weights are binary: one on that exact support and zero influence otherwise. No inverse-noise or other numerical weighting, cross-segment or cross-group borrowing, or gap bridging is permitted. The state includes support, mask, segment boundary, unit, denominator, and failure rule used to define $C_{\lambda_g}(Y_g) = Y_g - \lambda_g$. The location is discarded rather than restored because detector x has no scientifically meaningful optical DC response.

`SCI-PTC-DEF-008` – **Scaling state.** The declared nonzero internal scale, axis, population, weights, estimator, support, unit, and inverse transform used for numerical conditioning.

`SCI-PTC-DEF-009` – **Basis-fit support.** Detector-time occurrences permitted to influence construction or selection of a temporal basis or supplied-template fit.

`SCI-PTC-DEF-010` – **Loading-fit support.** Occurrences permitted to influence the detector loading of an already declared temporal basis or template.

`SCI-PTC-DEF-011` – **Application support.** Occurrences to which the exact frozen application map is authorized to apply, including an explicitly allowed fit-excluded/apply-allowed state only where the resolved operator defines the value from admitted frozen state and support.

`SCI-PTC-DEF-012` – **Output support.** Occurrences eligible to remain in a transformed output; it is independent of fit and application supports.

`SCI-PTC-DEF-013` – **Cause and named-use support policy.** Producer facts and causes accumulate without erasure in $\mathcal{C}$. For each exact PTC-owned scientific use U, an immutable owner-controlled policy supplies the complete required-restriction set $\mathcal{R}_U$, exact inputs, T/F/U/C knowledge states, exceptions, scope, use, missing/conflict behavior, and policy/version identity. VAL evaluates the supplied proposition under the common total knowledge rule but does not invent PTC science. A downstream use owner applies its own rule to the preserved facts supplied by PTC.

`SCI-PTC-DEF-014` – **Fixed-state application map.** The exact group-local numerical map $\mathcal{A}_{\Theta_g}$ obtained after all fitted, selected, classified, ranked, grouped, masked, metric, support, coordinate, centering location λ_g, generalized-inverse, tolerance, and failure state is frozen. It subtracts but does not restore the frozen λ_g. For the ordinary PCA/SVD family it recomputes time-local coefficients against the frozen detector-loading subspace using the exact group-local application-influence state.

`SCI-PTC-DEF-015` – **Full PTC procedure.** The complete learn, fit, select, rank, classify, refit, apply, and publish operation beginning from the immutable admitted CAL parent.

`SCI-PTC-DEF-016` – **Response companion.** An exact perturbation with a declared domain and parent, carried through the exact fixed-state derivative or used in a full-procedure injection study. A detector-time companion already on the CAL grid is not passed through $K_{\mathrm{up}\to\mathrm{CAL}}$ again, and no companion silently enters the fit.

`SCI-PTC-DEF-017` – **Conditional response.** The derivative or declared affine linear part of the frozen application map, conditioned on the exact resolved selected model. It excludes movement of learned, selected, classified, or fallback state.

`SCI-PTC-DEF-018` – **Full-procedure response.** A finite-perturbation signal response produced by rerunning the complete PTC procedure from the immutable admitted CAL parent. It is paired with a typed comparison of changed discrete/mixed state and may be a family over amplitude, side, source state, or location.

`SCI-PTC-DEF-019` – **Post-fit detector assessment.** A bounded evaluation of declared residual, loading, influence, stability, source-response, and diagnostic quantities. In the ordinary configured-rank route it is advisory and cannot change group membership, support, rank, or resolved state.

`SCI-PTC-DEF-020` – **Support-changing refinement.** A separately authorized future capability in which a post-fit classification change alters basis-fit or loading-fit support and therefore requires one complete refit from the immutable admitted CAL parent. The ordinary configured-rank route permits zero such refinements.

`SCI-PTC-DEF-021` – **PTC pass.** A complete PTC realization with an immutable input parent and distinct realized identity. It is neither an internal solver iteration nor an external FRUIT recurrence.

`SCI-PTC-DEF-022` – **Diagnostic-only conditioned r.** A separately authorized, PTC-compatible, electronics-enriched r product used only for inert or advisory diagnostics in base v0.1. It cannot control or subtract calibrated x.

`SCI-PTC-DEF-023` – **Fitted loading.** A gauge-dependent coefficient coupling a temporal basis/template to one detector. It is not a gridding weight, precision, significance, or sensitivity.

`SCI-PTC-DEF-024` – **Analysis/gridding coefficient.** An explicitly named detector-indexed or detector-time-indexed coefficient permitted for downstream aggregation. Its index family, identity, unit, normalization operator and domain, factors, support, and lifecycle are declared; it is nonprecision unless separately proven.

`SCI-PTC-DEF-025` – **Typed availability.** A role-specific response, covariance, coefficient, or diagnostic status among `computed_published`, `not_computed_or_not_requested_for_this_product`, `invalid`, or `unavailable`, with a cause and affected claim. It is distinct from whether a product exists or was realized.

`SCI-PTC-DEF-026` – **Least-aggressive admissible candidate.** For a separately requested future automatic-selection family, the first candidate under a declared ordering for which every required residual-contamination, astronomical-transfer, conditioning, support, stability, and QC predicate passes. No failed predicate can be offset by a scalar score. This is not the ordinary configured-rank route.

`SCI-PTC-DEF-027` – **Map-center response diagnostic.** The optional scalar functional of an exact source companion, propagated PTC response, exact named reference MAP operator, and center selector. It is neither the general PTC response nor MAP authority.

`SCI-PTC-DEF-028` – **Material state.** Every requested, resolved, learned, fitted, selected, classified, applied, randomized, fallback, or publication quantity capable of changing a declared output, response, coefficient, uncertainty, or downstream use.

`SCI-PTC-DEF-029` – **Learned/fitted evidence state.** The complete record $\mathfrak{L}$ of every evaluated candidate-model specification and its fitted or failed state. For the ordinary route it retains the one explicitly requested positive-rank candidate, exact group-local feasibility evidence, diagnostics, and cause. A future automatic-selection family additionally retains every predicate value and unit, pass/fail cause, ordering, tie result, and disposition.

`SCI-PTC-DEF-030` – **Resolved selected model.** The immutable group-local state Θ_g resolved from learned evidence, including its configured network- or array-level identity, exact basis/subspace, detector-right acting space, metric, mask-aware coefficient-recomputation rule, supports, coordinates, strictly positive requested rank, degeneracy, generalized-inverse/tolerance, classes, failure state, and application map.

`SCI-PTC-DEF-031` – **Frozen-subspace projection family.** An admitted PCA/SVD application family that recomputes application coefficients linearly against a frozen realized subspace and metric. The ordinary configured-rank route uses group-local detector-right, mask-aware coefficient recomputation. Any temporal-left, two-sided, detector/time-specific, or general vectorized action is a separately named family rather than an inferred orientation.

`SCI-PTC-DEF-032` – **Frozen-component subtraction family.** A separately named affine family $Y' \mapsto Y' - \widehat{U}_{\text{total}}$ with the typed frozen operand $\widehat{U}_{\text{total}} := U_{\text{total},\Theta}(Y^{\text{CAL}})$. Its derivative is the identity; it is not interchangeable with frozen-subspace projection and is not the ordinary PCA/SVD application rule.

`SCI-PTC-DEF-033` – **Realized propagated companion.** The detector-time companion after exactly one compatible application of the requested local or chain response. It records whether its parent was source-domain or already on the CAL grid and whether propagation was conditional or full-procedure.

`SCI-PTC-DEF-034` – **Typed state comparison.** A field-aware comparison Δ_{state} of learned, selected, classified, ranked, grouped, support, convergence, and fallback state across two procedure runs. Discrete state is compared by identity and disposition, never numerically subtracted.

`SCI-PTC-DEF-035` – **Product realization state.** A state among `realized`, `disabled`, `not_requested`, `rejected`, or `unavailable` for a named product role after its producing stage is entered. It is orthogonal to response or uncertainty availability. The RTC-terminal export route does not enter CAL or PTC and therefore does not fabricate PTC product-role states.

`SCI-PTC-DEF-036` – **External source/residual parent.** A source model, mask, residual, or prior-pass parent admitted only with explicit owner, unit, coordinate frame, registration, validity, response, support, and recurrence identity.

`SCI-PTC-DEF-037` – **Shifted/null surrogate.** A derived realization in which signal, associated producer facts, causes, support decisions, boundaries, randomness, and parent identity are transformed together under one declared shift or null rule.

`SCI-PTC-DEF-038` – **Astronomical-transfer predicate.** A candidate-admission predicate bound to an exact source class and companion, conditional or full-procedure response type, amplitude/location domain, metric, tolerance, mask/support, upstream response identity, and response-uncertainty treatment.

`SCI-PTC-DEF-039` – **Expectation/bias claim.** A family- and role-specific statement about the conditional expectation or bias of astronomical signal, fitted nuisance, remaining detector noise, calibration uncertainty, and selection. A missing or unproved claim is explicitly unavailable rather than silently unbiased.

`SCI-PTC-DEF-040` – **Application-available occurrence.** An occurrence, including a fit-excluded occurrence, for which the resolved frozen operator has every required detector binding, basis coefficient or coefficient-recomputation input, neighboring support, metric term, coordinate transform, and boundary state needed to define the application value without interpolation, zero filling, or an inferred extension.

`SCI-PTC-DEF-041` – **Complete upstream response ending on the CAL grid.** The exact admitted response $K_{\mathrm{up}\to\mathrm{CAL}}$ carried by the CAL product from a declared source-domain perturbation to the CAL detector-time grid. It composes every applicable admitted source-to-detector relation, beam convention, coordinate and scan geometry, RTC operation, and CAL operation; omission or unavailability of a required factor makes the complete-response claim unavailable.

`SCI-PTC-DEF-042` – **Ordinary configured-rank PCA route.** The requested PTC route that selects exactly one network- or array-level grouping, detector-wise time-axis centering, identity internal scaling, group-local mask-aware detector-space PCA/SVD, one explicitly configured strictly positive rank, one immutable-parent fit per group, and zero support-changing refinements.

`SCI-PTC-DEF-043` – **Group-local operator state.** The exact detector membership, segment, fit, loading, application, and output supports, influence masks, centering, metric, fitted loading subspace, strictly positive rank, degeneracy, generalized inverse and tolerance, coefficient-recomputation rule, fixed-state kernel operator, failure state, and provenance for one configured network or array group.

`SCI-PTC-DEF-044` – **RTC-terminal export route.** The successful workflow selected only by an explicit PTC-disabled request: publish the complete required consumer-neutral RTC bundle and terminate Citlali after RTC without entering CAL, PTC, or MAP. Failure of the required RTC publication fails the route.

`SCI-PTC-DEF-045` – **Fail-closed rank resolution.** The rule that rank zero, a noninteger rank, or a positive rank not realizable on the exact group-local support produces no PTC transformed signal and no map, with exact cause and without rank clipping, substitution, centering-only output, PTC-disabled reinterpretation, or RTC-export fallback.

7 Normative equations

The organizing detector-time model separates the latent correlated component from its fitted estimate:

$$Y^{\mathrm{CAL}}_{td} = S_{td} + U_{\star,td} + N_{td}, \qquad U_{\star,td} = \sum_{k=1}^{K} M_{\star,tk} A_{\star,dk}. \tag{1}$$

For the immutable fitted parent, distinguish the scaled-coordinate correlated fit, its physical-unit realization, the operator-primary output, and the total removed signal:

$$\widehat{\widehat{U}}_{\mathrm{corr}} = \widehat{M}\widehat{A}^{\mathsf{T}}, \qquad \widehat{U}_{\mathrm{corr}} = D\widehat{\widehat{U}}_{\mathrm{corr}}, \qquad Z = \mathcal{A}_{\Theta}(Y^{\mathrm{CAL}}), \qquad U_{\mathrm{total},\Theta}(Y^{\mathrm{CAL}}) = Y^{\mathrm{CAL}} - Z. \tag{2}$$

Equation 2 does not identify either removed quantity with one physical contaminant. The exact application operator defines the result; a fitted correlated component alone does not define how a new input or response companion is acted on.

For the ordinary route, let $\mathcal{T}^{\mathrm{ctr}}_{g,d}$ contain exactly the finite basis-fit-admitted time occurrences for detector d in one immutable PTC segment and configured group g. Its binary centering influence is one on that support and zero otherwise. The learned location, fitting coordinate, and physical-unit application map are

$$\begin{aligned}\lambda_{g,d} &= \frac{\displaystyle\sum_{t\in\mathcal{T}^{\mathrm{ctr}}_{g,d}} w^{\mathrm{ctr}}_{gtd} Y^{\mathrm{CAL}}_{gtd}}{\displaystyle\sum_{t\in\mathcal{T}^{\mathrm{ctr}}_{g,d}} w^{\mathrm{ctr}}_{gtd}}, \qquad w^{\mathrm{ctr}}_{gtd} = 1 \quad (t \in \mathcal{T}^{\mathrm{ctr}}_{g,d}),\\ Y'_c &= C_{\lambda_g}(Y') = Y' - \lambda_g, \qquad \widetilde{Y}' = D^{-1}Y'_c,\\ \mathcal{A}_{\Theta_g}(Y') &= D\widetilde{\mathcal{A}}_{\Theta_g}(\widetilde{Y}').\end{aligned} \tag{3}$$

The denominator must be finite and strictly positive; otherwise the affected group fails closed. No occurrence outside the exact segment and support enters the sum, including across gaps, networks, or arrays. Ordinary internal scaling is identity. Any separately declared invertible scaling is restored for ordinary output; λ is not. Detector x has no scientifically meaningful optical DC response, so the removed additive reference is discarded and published in $\mathcal{N}_{\mathrm{PTC}}$. In fixed-state application λ is held fixed. In a full-procedure response it is re-estimated in each run, may move under perturbation, and is again discarded.

For the linear specialization $\widetilde{\mathcal{A}}_\Theta(\widetilde{Y}') = L_\Theta \widetilde{Y}'$, the correlated-subspace and total removed signals are

$$U_{\mathrm{corr},\Theta}(Y') = D(I - L_\Theta)D^{-1}(Y' - \lambda), \qquad U_{\mathrm{total},\Theta}(Y') = \lambda + U_{\mathrm{corr},\Theta}(Y').$$

Facts and causes accumulate without erasure, while the exact named-use owner supplies a complete required-restriction set evaluated under common T/F/U/C knowledge semantics:

$$\begin{aligned}\mathcal{C} &= \bigcup_i \mathcal{C}_i,\\ \mathrm{Decision}_U &= \begin{cases}\texttt{ineligible}, & \exists r \in \mathcal{R}_U:\ K(r) = \mathsf{F},\\ \texttt{eligible}, & \forall r \in \mathcal{R}_U:\ K(r) = \mathsf{T},\\ \texttt{decision\_unavailable}, & \text{otherwise.}\end{cases}\end{aligned} \tag{4}$$

This rule follows exact request, structural binding, and applicability checks. Known inapplicability yields no eligibility proposition. One applicable false restriction cannot be rescued by another cause; all causes, unknowns, and conflicts remain visible. PTC owns each PTC-use proposition, VAL evaluates the supplied rule, and another named-use owner supplies its own proposition while consuming the preserved facts.

An admitted group-local weighted or robust low-rank family minimizes

$$(\widehat{M}_g, \widehat{A}_g) = \arg\min_{M,A} \sum_{t,d} w^{\mathrm{fit}}_{td}\, \rho\Big(\widetilde{Y}_{td} - (MA^{\mathsf{T}})_{td}\Big), \tag{5}$$

where $w^{\mathrm{fit}}_{td} = 0$ denotes no fit influence, not a zero-valued observation. The loss, metric, support, rank, and missing-data semantics are declared.

For a fixed supplied design B and fit metric Ω, a conditioned regression family is

$$\widehat{c} = (B^{\mathsf{T}}\Omega B)^{+}B^{\mathsf{T}}\Omega y, \qquad \mathcal{L}_{B,\Omega} = I - B(B^{\mathsf{T}}\Omega B)^{+}B^{\mathsf{T}}\Omega. \tag{6}$$

The generalized inverse, tolerance, units, gauge, coefficient support, and application support are material state.

A weighted common-mode family may estimate its temporal mode as

$$\widehat{m}_t = \frac{\sum_d w^{\mathrm{fit}}_{td}\omega^{\mathrm{cm}}_{td}\widetilde{Y}_{td}}{\sum_d w^{\mathrm{fit}}_{td}\omega^{\mathrm{cm}}_{td}}, \tag{7}$$

only where the denominator is finite and nonzero. The aggregation weight $\omega^{\mathrm{cm}}_{td}$, its unit, normalization, and population are declared; it is not a fitted detector loading by implication.

For the ordinary configured-rank PCA/SVD route, application uses the frozen group-local detector-loading subspace and mask-aware linear coefficient recomputation. For centered row $x_{g,t} = Y^{\mathrm{CAL}}_{g,t} - \lambda_g$,

$$\begin{aligned}\widehat{m}^{\mathrm{app}}_{g,t} &= x_{g,t}W^{\mathrm{app}}_{g,t}\widehat{A}_g\left(\widehat{A}^{\mathsf{T}}_g W^{\mathrm{app}}_{g,t}\widehat{A}_g\right)^{+},\\ z_{g,t} &= x_{g,t} - \widehat{m}^{\mathrm{app}}_{g,t}\widehat{A}^{\mathsf{T}}_g.\end{aligned} \tag{8}$$

The exact output support determines which entries of $z_{g,t}$ are published. The generalized inverse, tolerance, detector binding, metric, support, and insufficient-support state are frozen in Θ_g. On complete uniform support this reduces to detector-right projection through $I - \widehat{A}_g(\widehat{A}_g^\top \widehat{A}_g)^+ \widehat{A}_g^\top$. Any temporal-left, two-sided, detector/time-specific, or general vectorized family remains separately named and is not inferred from matrix orientation or rank.

A separately named frozen-component subtraction family is

$$\widehat{U}_{\text{total}} := U_{\text{total},\Theta}(Y^{\text{CAL}}), \qquad \mathcal{A}^{\text{sub}}_{\widehat{U}_{\text{total}}}(Y') = Y' - \widehat{U}_{\text{total}}, \qquad D\mathcal{A}^{\text{sub}}_{\widehat{U}_{\text{total}}}[Y](H) = H. \tag{9}$$

It is not interchangeable with Equation 8 and is not the base-v0.1 PCA/SVD application rule.

The ordinary route selects exactly one configured grouping level:

$$\mathcal{G} = \begin{cases} \{D_a\}, & \text{array-level PCA,} \\ \{D_{a,n}\}_n, & \text{network-level PCA.} \end{cases} \tag{10}$$

Each $g \in \mathcal{G}$ has independent group-local support and resolved state. The alternatives are not automatically sequential. A future hierarchical family requires separate authority for its exact joint or sequential order.

For the ordinary route, the finite candidate-model set contains exactly the explicitly requested positive-rank model specification:

$$\mathcal{M}_{\text{cand},g} = \{c_g(k_{\text{req},g})\}, \qquad k_{\text{req},g} \in \mathbb{Z}, \qquad 1 \le k_{\text{req},g} \le k_{\text{admissible},g}. \tag{11}$$

The first route performs mathematical feasibility resolution, not automatic scientific rank selection. Rank zero and unrealizable positive rank fail with cause and no fallback, clipping, centering-only output, route conversion, or map. A future automatic-selection family may use a separately authorized conjunctive least-aggressive rule and must retain every predicate, threshold, uncertainty treatment, order, tie, and rejected candidate in $\mathfrak{L}$.

The ordinary route performs one immutable-parent fit per configured group and zero support-changing refinements. For a separately authorized future refinement family, the immutable-parent lifecycle is

$$\begin{aligned} \Theta_j &= \text{Fit}(Y^{\text{CAL}}; \mathcal{S}_j), \\ \mathcal{S}_{j+1} &= \text{Classify}(Y^{\text{CAL}}, \Theta_j), \\ \Theta_{j+1} &= \text{Fit}(Y^{\text{CAL}}; \mathcal{S}_{j+1}). \end{aligned} \tag{12}$$

Every fit uses the same immutable admitted CAL parent. A cleaned Z_j is not the next fit parent, and the final complete model is applied once.

The one-way state lifecycle is

$$R \longrightarrow E \longrightarrow O \longrightarrow \mathfrak{L} \longrightarrow \Theta \longrightarrow \Lambda \longrightarrow \Pi, \tag{13}$$

where learned evidence, resolved selection, numerical application, and publication are distinct immutable states.

The full PTC procedure has separately typed signal and state projections:

$$\mathcal{P}(Y^{\text{CAL}}, R, E, O) = \big(\mathcal{P}_Z(Y^{\text{CAL}}), \mathcal{P}_\Lambda(Y^{\text{CAL}})\big) = (Z, \Lambda). \tag{14}$$

For fixed group-local resolved model Θ_g, including frozen λ_g, the local detector-time response and a realized CAL-grid companion are

$$J_{\Theta_g}[Y_g] = D\mathcal{A}_{\Theta_g}[Y_g], \qquad k^{\text{PTC}|\Theta_g}_{\text{src},g} = J_{\Theta_g}[Y_g]\, k^{\text{CAL}}_{\text{src},g}. \tag{15}$$

The companion uses the same group, support, mask, metric, rank, loading subspace, generalized inverse, and tolerance as the data. The frozen λ_g is not subtracted from a perturbation and is not re-estimated. If the parent is source-domain, the complete chain response is

$$k^{\text{CAL}}_{\text{src}} = K_{\text{up}\to\text{CAL}}\tau_{\text{src}}, \qquad K_{\text{PTC}|\Theta} = J_\Theta[Y] \circ K_{\text{up}\to\text{CAL}}, \qquad k^{\text{PTC}|\Theta}_{\text{src}} = K_{\text{PTC}|\Theta}\tau_{\text{src}}. \tag{16}$$

Equations 15–16 are alternative entry points for different companion parents; $K_{\text{up}\to\text{CAL}}$ is never applied twice.

For a declared detector-time perturbation h^{CAL}, the full-procedure finite-perturbation signal response is

$$K^{\text{proc}}_{\text{PTC}}(\epsilon, h^{\text{CAL}}) = \frac{\mathcal{P}_Z(Y^{\text{CAL}} + \epsilon h^{\text{CAL}}) - \mathcal{P}_Z(Y^{\text{CAL}})}{\epsilon}. \tag{17}$$

Each full-procedure evaluation re-estimates and again discards λ. The corresponding state evidence, including any centering-state movement, is retained separately:

$$\Delta_{\text{state}}(\epsilon, h^{\text{CAL}}) = \text{Compare}_{\text{typed}}\left[\mathcal{P}_\Lambda(Y^{\text{CAL}} + \epsilon h^{\text{CAL}}), \mathcal{P}_\Lambda(Y^{\text{CAL}})\right]. \tag{18}$$

Amplitude, side, source state, location, domain, and every changed rank, mask, group, support, class, convergence, or fallback field are part of the result.

For response role $r \in \{\text{fixed}, \text{procedure}\}$, the optional map-center diagnostic is

$$\rho^{r}_{\text{center}} = e^{\mathsf{T}}_{\text{center}} A_{\text{ref}} k^{\text{PTC,r}}_{\text{src}}. \tag{19}$$

The exact response role, source-companion parent, reference MAP operator, and center selector are mandatory factors. The diagnostic confers no MAP or general beam authority.

For a fixed differentiable realized map, conditional covariance is

$$C_{Z|\Theta} = J_{\Theta}[Y]\, C_Y\, J_{\Theta}[Y]^{\mathsf{T}}. \tag{20}$$

For random selected state Θ, the exact decomposition is

$$\text{Cov}(Z) = \text{E}[\text{Cov}(Z \mid \Theta)] + \text{Cov}(\text{E}[Z \mid \Theta])\,. \tag{21}$$

Omitting the second term is a conditioned approximation, not complete uncertainty.

A diagnostic indexed by detector or detector-time occurrence may take the generic residual form

$$\eta_i = H_i(Z, \widehat{U}_{\text{corr}}, U_{\text{total},\Theta}, \mathcal{S}, \Theta; \chi_H), \tag{22}$$

where χ_H records the domain, population/model reference, normalization, support, uncertainty, and policy role.

A downstream analysis/gridding coefficient family is

$$\widetilde{\gamma}_i = f_{\gamma}(\eta_i; \chi_{\gamma}), \qquad \gamma_i = \mathcal{N}_{\gamma}(\widetilde{\gamma}_i; \{\widetilde{\gamma}_j : j \in \mathcal{D}_{\gamma}\})\,, \tag{23}$$

where $i = d$ or $i = (t, d)$ is declared, $\mathcal{D}_{\gamma}$ is the exact normalization domain, and f_{γ} and $\mathcal{N}_{\gamma}$ are named operators. The result is not precision unless separately proved.

Transitive causal influence for output j is

$$\mathcal{I}_j = \bigcup_{i:\, i \to j} (\{i\} \cup \mathcal{I}_i)\,, \tag{24}$$

including inputs that affect fitted state or selection even when a local numeric derivative vanishes.

An explicit PTC-disabled request selects the RTC-terminal export route:

$$\mathsf{PTC\ request} = \mathsf{disabled} \Longrightarrow \mathsf{route} = \mathsf{RTC\_terminal\_export} \Longrightarrow \mathsf{terminal} = \mathsf{successful\ after\ complete\ RTC\ publication.} \tag{25}$$

CAL, PTC, and MAP are not entered and no products for those stages are realized. Failure of the required RTC publication fails the route. Rank zero is not an alias for this transition, and no direct CAL-to-MAP fallback exists for ordinary Citlali reduction.

8 Normative assumptions and declared limits

`SCI-PTC-ASM-001`. The primary numerical parent is an admitted SCI-CAL calibrated x product in the declared top-of-atmosphere, point-source-equivalent mJy-per-fixed-nominal-beam convention.

`SCI-PTC-ASM-002`. The complete RTC and CAL lineage, including exact detector/time identities, validity, influence, response, and uncertainty status, is available or the affected claim fails closed.

`SCI-PTC-ASM-003`. Correlation alone does not identify atmosphere, electronics, calibration residual, detector noise, astronomical signal, or any unique mixture.

`SCI-PTC-ASM-004`. PTC fitting is confined to one array unless a successor authority supplies a cross-array spectral, calibration, beam, source-color, and response model.

`SCI-PTC-ASM-005`. The selected estimator family supplies explicit missing-data semantics. Replacing a missing or excluded value with numerical zero is not exclusion unless the estimator proves equivalence.

`SCI-PTC-ASM-006`. Every centering/scaling transform has a declared axis, population, support, weights, estimator, boundary, unit, gauge, failure behavior, and null-space consequence. Centering subtracts a learned additive location λ that is not restored because detector x has no scientifically meaningful optical DC response.

`SCI-PTC-ASM-007.` Internal scaling is invertible and restored before ordinary transformed output. The learned additive location is not restored. Any other noninvertible operation changes the signal definition and requires a separately named output role.

`SCI-PTC-ASM-008.` A source mask protects only its declared model and registered support. Unmasked compact or extended sky components may overlap the fitted subspace.

`SCI-PTC-ASM-009.` A separately requested future automatic-selection family is finite, conjunctive, deterministically ordered, and least-aggressive. Numerical predicates are owner-controlled inputs rather than inferred defaults. The ordinary route instead uses one explicitly requested positive-rank candidate.

`SCI-PTC-ASM-010.` The ordinary route performs zero support-changing detector refinements. Any separately authorized future support-changing refinement refits the complete selected model from the same immutable admitted CAL parent.

`SCI-PTC-ASM-011.` Network- and array-level PCA are alternative configured ordinary routes rather than sequential stages. A future sequential residual fit is valid only when declared as part of one complete ordered hierarchical estimator with cumulative removed subspace, response, covariance, and parentage.

`SCI-PTC-ASM-012.` The conditioned-r branch, when available, has its own operator, unit, response, validity, optical-leakage state, and exact relation to the calibrated-x grid.

`SCI-PTC-ASM-013.` Base-v0.1 r analysis is inert or advisory. It cannot change calibrated-x fit membership, requested or resolved rank, subtraction, output retention, or coefficients and cannot supply an x subtraction basis.

`SCI-PTC-ASM-014.` A fixed-state application map does not by itself represent the response of data-derived rank, mask, grouping, clipping, classification, convergence, or fallback decisions.

`SCI-PTC-ASM-015.` A whole-chain RTC-to-CAL-to-PTC injection is distinct from the PTC full-procedure response and requires separately named upstream owners and response companions.

`SCI-PTC-ASM-016.` Full covariance is optional. An absent covariance or cross-covariance is unknown/unavailable rather than zero, independent, or diagonal.

`SCI-PTC-ASM-017.` Fitted loadings, centering/scaling parameters, diagnostic quantities, and analysis/gridding coefficients are distinct types and lifecycles.

`SCI-PTC-ASM-018.` A scalar analysis/gridding coefficient is nonprecision unless its exact target variance, conditioning, unit, normalization, covariance, and independence conditions are proven.

`SCI-PTC-ASM-019.` PTC internal iteration, a new PTC pass, and an external FRUIT recurrence are distinct identities with distinct parents and stop rules.

`SCI-PTC-ASM-020.` The in-memory transformed TOD is an authoritative intermediate for the PTC-dependent route, not an independent sky estimator or a claim of map, beam, significance, or production validity.

`SCI-PTC-ASM-021.` A required unavailable input, response term, support, identity, or output makes only the affected product or claim unavailable; it is never replaced silently by a numeric default.

`SCI-PTC-ASM-022.` Implementation conformity, representation fidelity, observational performance, and production readiness require separate evidence and are not established by this contract.

`SCI-PTC-ASM-023.` For every admitted base-v0.1 PCA/SVD family, application to science or a fixed-state companion is projection through the frozen realized subspace and metric with linear coefficient recomputation. The ordinary route uses group-local detector-right, mask-aware recomputation. Subtraction of one frozen fitted numerical component is a distinct family and is not inferred.

`SCI-PTC-ASM-024.` The ordinary route's detector-right action is explicit owner-selected state, not an inference from rank or matrix orientation. Any temporal-left, two-sided, detector/time-specific, or vectorized action is a separately named family with its exact acting space and operator.

`SCI-PTC-ASM-025.` Source-domain templates, CAL-grid detector-time companions, PTC-local operators, complete chain responses, and realized propagated companions are distinct typed objects. An upstream response operator is applied exactly once.

`SCI-PTC-ASM-026.` Learned candidate evidence, resolved selected model, applied state, and published product state are distinct immutable lifecycle stages even when one implementation stores them together.

`SCI-PTC-ASM-027.` An external source/residual parent or shifted/null surrogate is scientifically usable only when its complete parent, coordinate, validity, support, response, recurrence, and transformation state is available; insufficient support is not a valid zero surrogate.

`SCI-PTC-ASM-028.` Fit exclusion does not itself authorize application. A fit-excluded occurrence is application-available only when the resolved frozen operator defines that occurrence from admitted frozen state and support without inferred interpolation, zero filling, or extension.

`SCI-PTC-ASM-029.` Producer facts and causes accumulate without erasure. PTC owns complete policy propositions for PTC-local scientific uses; the owner of any other named use owns its admission policy. Shared VAL machinery represents T/F/U/C knowledge state, preserves causes, evaluates a supplied total rule, and records provenance, but it does not invent package-specific scientific predicates. Any decisive false yields ineligible; all true yields eligible; otherwise an applicable requested use is decision unavailable.

`SCI-PTC-ASM-030.` The ordinary route selects exactly one configured grouping level: one full array or the set of readout networks within one array. Every support, mask, centering state, fitted subspace, rank, application, and kernel response is scoped to that same group identity.

`SCI-PTC-ASM-031.` The ordinary route uses the detector-wise support-normalized arithmetic mean over finite basis-fit-admitted occurrences within each exact immutable PTC segment, with binary centering influence and identity internal scaling. It does not use inverse-noise or other numerical centering weights, borrow across segments or configured groups, or bridge gaps. The learned location is not restored, and no later baseline or FRUIT recurrence treatment is inferred.

`SCI-PTC-ASM-032.` The ordinary route rank is an explicitly requested integer greater than or equal to one. Rank zero and unrealizable positive rank fail closed and are not defaults, fallbacks, centering-only outputs, PTC-disabled states, or RTC-export requests.

`SCI-PTC-ASM-033.` Masked or excluded occurrences have zero statistical influence only through the declared group-local estimator and application masks; they are not numerical zero observations.

`SCI-PTC-ASM-034.` Post-fit diagnostics in the ordinary route are advisory and cannot change fitted support, grouping, rank, resolved model, or output application state.

`SCI-PTC-ASM-035.` An explicit PTC-disabled request selects a successful RTC-terminal export only after the complete required RTC bundle is published. CAL, PTC, and MAP are not entered; failure of required RTC publication fails the route.

9 Normative requirements

9.1 Input identity, lineage, and state

`SCI-PTC-REQ-001` – **Admitted calibrated signal.** PTC shall accept as its primary numerical parent only an admitted sample-by-detector SCI-CAL x product in top-of-atmosphere, point-source-equivalent mJy per fixed nominal beam, with explicit response state.

`SCI-PTC-REQ-002` – **Complete RTC lineage.** The admitted bundle shall retain conditioned-x identity, raw-r parent identity, exact time grid, and every causal RTC selector, segmentation, validity, response, uncertainty, replacement, and influence state.

`SCI-PTC-REQ-003` – **Upstream availability.** An invalid or unavailable required RTC or CAL fact shall make the affected calibrated PTC product or claim unavailable; PTC shall not repair, approximate, or relabel it.

`SCI-PTC-REQ-004` – **Matrix shape.** Every primary matrix shall declare samples on rows, detectors on columns, actual extents, flattening/storage order, and the bijection between positions and scientific identities.

`SCI-PTC-REQ-005` – **Time identity.** Sample time shall carry declared scale, epoch, unit, actual timestamps, gaps, segment boundaries, and any cross-boundary support; nominal rate shall not replace actual time identity.

`SCI-PTC-REQ-006` – **Detector identity.** Detector joins shall use immutable occurrence/UID identity and exact APT binding; dense columns, file order, or local integers shall not be external identities.

`SCI-PTC-REQ-007` – **Within-array domain.** Every base fit shall be confined to one declared array. Cross-array modes shall be unavailable without separately authorized spectral, calibration, beam, source-color, and response authority.

`SCI-PTC-REQ-008` – **Requested operation.** The requested state shall identify estimator family, configured network- or array-level grouping, strictly positive rank, centering/scaling, gauge, group-local support and masks, refinement, source protection, coefficient families, response, covariance, simulation, and output roles; absence shall differ from an explicit value. Candidate ordering and conjunctive predicates are required only for a separately requested automatic-selection family.

`SCI-PTC-REQ-009` – **One-way state lifecycle.** Requested, effective-policy, observation-resolved input, learned/fitted evidence, resolved selected model, applied/realized state, and published product state shall be immutable, one-way linked, and distinct as in Equation 13. A later stage shall not overwrite or reconstruct an earlier stage.

`SCI-PTC-REQ-010` – **Unit preservation limit.** Ordinary transformed x shall retain the admitted CAL unit and calibration convention, but unit retention shall not be labeled preservation of point-source peak, absolute level, extended-source response, detector combinations, or beam shape.

9.2 Validity, causes, and support

`SCI-PTC-REQ-011` – **Cause is not action.** Every flag shall remain a typed cause. Causes from all applicable producers shall accumulate without erasure; no cause name alone shall imply zero-fill, universal rejection, zero coefficient, refit, or downstream ineligibility.

`SCI-PTC-REQ-012` – **Complete named-use decision.** For each distinct PTC-owned scientific use, including basis fit, loading fit, operator application, output retention, coefficient/QC, response, empirical estimation, and simulation, the owner shall supply a complete required-restriction set with exact inputs, T/F/U/C knowledge, exceptions, missing/conflict behavior, scope, use, and policy/version identity. After request, structural binding, and applicability checks, any decisive false shall yield `ineligible`, all true shall yield `eligible`, and every remaining applicable requested state shall yield `decision_unavailable`. All facts and causes remain visible; permission for one use never implies permission for another. VAL evaluates the supplied rule and does not author PTC policy.

`SCI-PTC-REQ-013` – **Finite eligible arithmetic.** Only occurrences admitted for the exact PTC-owned use and finite under its numerical contract shall enter a fit, normalization, covariance estimate, convergence statistic, or coefficient statistic. Missing, non-finite, invalid, rejected, unavailable, `decision_unavailable`, and numeric zero shall remain distinct.

`SCI-PTC-REQ-014` – **No silent zero filling.** A missing or fit-excluded value shall not be replaced by numerical zero before PCA/SVD or another estimator unless exact equivalence to zero statistical influence is proved for the selected estimator.

`SCI-PTC-REQ-015` – **Fit-excluded apply-allowed state.** The support model shall express an occurrence excluded from mode or loading learning but eligible for application and output, without treating that state as fit validity.

`SCI-PTC-REQ-016` – **Direct synthesized/replaced occurrences.** Direct ALIGN-synthesized or RTC-replaced occurrences shall be excluded from the calibrated PTC science branch.

`SCI-PTC-REQ-017` – **Transitive influence.** Noncenter causal influence shall be preserved and evaluated under PTC's declared use-specific support and response policy; compact over-approximation is permitted and under-approximation is not.

`SCI-PTC-REQ-018` – **Mask identity.** Every mask shall declare polarity, use, parent, frame, unit, registration, support, boundary convention, and derivation state. One mask bit shall not silently govern every use.

9.3 Estimand, coordinates, and grouping

`SCI-PTC-REQ-019` – **Declared estimand.** Every realized operation shall identify the exact correlated-mode or supplied-template estimand, scaled-coordinate fit, physical-unit correlated component, operator-defined total removed signal, objective/loss, metric, parameterization, group, fit support, loading support, application support, acting space, and frozen-state application map.

`SCI-PTC-REQ-020` – **No physical-origin inference.** A fitted correlated component shall not be labeled atmosphere, electronics, calibration residual, detector noise, or sky solely because it is correlated.

`SCI-PTC-REQ-021` – **Removed subspace publication.** Every transformed product shall publish the complete removed subspace or template family, configured network- or array-level group, metric, strictly positive rank, support and masks, fit order, degeneracy, generalized-inverse state, and gauge-equivalence class. It shall distinguish $\widehat{U}_{\mathrm{corr}}$ from $U_{\mathrm{total},\Theta}$.

`SCI-PTC-REQ-022` – **Additive reference and null roles.** Every product shall publish additive-reference state and separately identify the fixed-state linear null, any re-estimated-centering invariant or unidentifiable modes, the affine additive-reference loss caused by discarding λ, and astronomical modes known or permitted to be attenuated.

`SCI-PTC-REQ-023` – **Centering contract.** Each centering operation shall declare configured group and immutable PTC segment, detector-wise time axis, population, support, weights, location estimator, masks, boundaries, units, gauge, denominator, failure state, and resulting null/loss roles. For the ordinary route, $\lambda_{g,d}$ shall be the support-normalized arithmetic mean over exactly the finite basis-fit-admitted occurrences of detector d in that segment and group, with binary centering influence. It shall use no inverse-noise or other numerical centering weight, no cross-segment or cross-group borrowing, and no gap bridging. The route shall subtract and publish λ_g but shall not restore it to conditioned detector x.

`SCI-PTC-REQ-024` – **Scaling contract.** Each scaling operation shall declare axis, population, support, weights, scale estimator, nonzero/finite rule, units, inverse, gauge, and failure state. The ordinary route shall use identity internal scaling; any other invertible internal scaling is a separately declared family and is restored before ordinary output.

`SCI-PTC-REQ-025` – **Gauge-invariant claim.** Scientific claims shall attach to the fitted correlated signal and removed subspace, not to unpinned basis sign, reciprocal basis/loading scale, component order, or rotation inside a degenerate subspace.

`SCI-PTC-REQ-026` – **Estimator-family identity.** The resolved selected model shall name the estimator family, its exact mathematical definition, and its distinct fit and application rules; robust common mode, fixed-template regression, masked/weighted PCA/SVD, frozen-component subtraction, and diagnostic-r PCA shall not be conflated.

`SCI-PTC-REQ-027` – **Robust common-mode family.** A common-mode family shall state the input coordinate, location estimator, aggregation weights and their unit/normalization, support, denominator rule, detector loading model, units, and robust-loss behavior; an aggregation weight shall not be labeled a fitted loading, and a median shall not be documented as an arithmetic mean.

`SCI-PTC-REQ-028` – **Fixed-template family.** A fixed-template regression shall state template parent, unit, gauge, metric, generalized inverse, tolerance, fit support, loading support, application support, and exact treatment of rank deficiency.

`SCI-PTC-REQ-029` – **Masked/weighted PCA family.** A PCA/SVD family shall state matrix orientation, configured group, the space containing the removed basis, every frozen quantity including λ_g, every coefficient recomputed linearly at application, centering/scaling, missing-data model, fit and application influence masks, weights/noise metric, candidate construction, component order, rank, degeneracy rule, generalized inverse/tolerance, fit support, and application support. The ordinary route shall use group-local detector-right mask-aware coefficient recomputation and shall act on a compatible kernel with the identical frozen state. It shall not subtract one frozen numerical reconstruction.

`SCI-PTC-REQ-030` – **Diagnostic-only conditioned r.** Conditioned-r PCA shall be available only with a separately authorized compatible parent and shall remain inert or advisory: it shall not alter calibrated-x membership, configured or resolved rank, subtraction, output, coefficients, or supply an x subtraction basis.

`SCI-PTC-REQ-031` – **Configured ordinary groups.** The ordinary route shall select exactly one grouping level: one array-wide group or the set of readout-network groups within one array. Every group shall identify exact detectors, coordinates, fit/loading/application/output supports, masks, centering, metric, rank, subspace, application, and kernel-response state.

`SCI-PTC-REQ-032` – **No implicit hierarchy.** Network- and array-level ordinary PCA shall be alternative configured routes and shall not be applied sequentially by implication. Any future overlapping hierarchical components shall declare joint fitting or exact sequential order and publish the cumulative removed subspace, response, covariance, and parentage.

`SCI-PTC-REQ-033` – **Data-derived grouping.** Data-derived or local/focal-plane grouping is unavailable for the ordinary route. A separately authorized future group shall record population, metric, candidate rule, cardinality controls, stability test, held-out or adjacent-segment status, selection uncertainty, and frozen application identity.

9.4 Mode selection and source protection

`SCI-PTC-REQ-034` – **Finite candidate set.** The ordinary route shall use the singleton candidate-model set containing its explicitly requested positive-rank model after centering, identity scaling, grouping, supports, and metric are fixed. A separately requested automatic-selection family shall evaluate its own finite declared candidate set.

`SCI-PTC-REQ-035` – **Conjunctive automatic admission.** Only a separately requested future automatic-selection family shall select the least aggressive candidate for which every required residual-contamination, astronomical-transfer, conditioning, support, stability, and QC predicate passes. Each predicate shall retain its typed value, unit, uncertainty treatment, threshold, comparison polarity, pass/fail cause, and claim. The ordinary configured-rank route performs no such automatic selection.

`SCI-PTC-REQ-036` – **No compensating score.** For any future automatic-selection family, failure of a required predicate shall not be offset by another predicate or scalar score. No score or diagnostic shall change the ordinary route's explicitly requested rank.

`SCI-PTC-REQ-037` – **Ordering and ties.** Candidate ordering, least-aggressive relation, deterministic tie handling, and boundary comparison polarity shall be declared before any future automatic selection. The singleton ordinary route has no data-derived ordering or tie decision.

`SCI-PTC-REQ-038` – **Nonnested candidates.** A future automatic-selection family shall compare nonnested candidates by complete removed subspace and astronomical response rather than mode count alone. The ordinary configured-rank route does not compare candidates.

`SCI-PTC-REQ-039` – **Failed candidate resolution.** If the ordinary requested positive-rank candidate is unrealizable on its exact group-local support, no transformed product or map shall be produced for the route; rank shall not be clipped, changed, or converted to another route. If no candidate passes a future automatic-selection family, its requested transformed product shall likewise be unavailable or rejected with cause and no silent fallback.

`SCI-PTC-REQ-040` – **No universal threshold.** No universal mode count, variance fraction, eigengap, or singular-value threshold shall be treated as scientific authority outside its declared model and evidence.

`SCI-PTC-REQ-041` – **Source-mask limit.** Source protection is not requested by the ordinary route. Any separately authorized source mask shall protect only its declared source model and registered support and shall not be represented as proof that unmasked compact emission, extended emission, or other sky modes survive.

9.5 Post-fit detector assessment and repetition

`SCI-PTC-REQ-042` – **Diagnostic definition.** Every residual, loading, influence, stability, source-response, non-Gaussianity, transient, or x/r diagnostic shall declare estimator, population or approved noise/signal reference, normalization, support, uncertainty, and policy role. Ordinary-route diagnostics are advisory and cannot alter the fitted state.

`SCI-PTC-REQ-043` – **Pathology discrimination.** Any post-fit assessment shall distinguish detector pathology from astronomical signal, source-model/mask failure, calibration state, focal-plane position, and expected detector-sensitivity variation. In the ordinary route this distinction is diagnostic only and initiates no reclassification or refit.

`SCI-PTC-REQ-044` – **Owner-controlled thresholds.** Numerical classification thresholds shall be explicit owner-controlled inputs. A missing threshold shall not be inferred from current data, code, or population behavior. No such threshold changes the ordinary route's configured rank or support.

`SCI-PTC-REQ-045` – **Finite refinement.** The ordinary route shall perform zero support-changing refinements. Any separately authorized fit-diagnose-classify-refit process shall be finite and shall record each support, diagnostic, class, refit decision, and emitted final state.

`SCI-PTC-REQ-046` – **Immutable-parent refit.** The ordinary route performs no fit-support-changing refinement. Every separately authorized future refinement shall refit one complete selected model from the same immutable admitted CAL parent and apply the final complete model once. A previously cleaned output shall not be the refit parent.

`SCI-PTC-REQ-047` – **Non-fit dispositions.** A postfit-output rejection, coefficient-only decision, or advisory diagnostic shall not retroactively change the fitted state. In the ordinary route no diagnostic may change support, grouping, rank, or Θ_g.

`SCI-PTC-REQ-048` – **Refinement stopping state.** The ordinary route stop state shall record one completed immutable-parent fit and zero refinements. A future refinement family shall distinguish stable support/subspace, no-new-threshold event, maximum refinements, oscillation, insufficient support, nonconvergence, exhaustion, and error; oscillation shall not be called convergence.

`SCI-PTC-REQ-049` – **Internal iteration identity.** Internal numerical iteration shall record statistic, norm/support, tolerance and unit, minimum/maximum iterations, comparison polarity, empty/non-finite behavior, selected iterate, and converged/exhausted/error status.

`SCI-PTC-REQ-050` – **PTC pass identity.** A new PTC pass shall have an immutable new input parent and distinct realized identity; it shall not overwrite a previous pass.

`SCI-PTC-REQ-051` – **External recurrence boundary.** Model subtraction/add-back, map feedback, external recurrence, restart, and recurrence parentage shall remain FRUIT or another declared recurrence owner's authority.

9.6 Coefficients and uncertainty

`SCI-PTC-REQ-052` – **Coefficient taxonomy.** Fitted loadings, centering/scaling parameters, diagnostic coefficients, and analysis/gridding coefficients shall be distinct typed families. Only an explicitly named analysis/gridding family may be MAP-facing.

`SCI-PTC-REQ-053` – **Fitted-loading semantics.** A fitted loading shall declare basis/template identity, gauge, unit, group, fit support, lifecycle, and numerical use and shall prohibit weight, precision, significance, sensitivity, or independent-noise interpretation.

`SCI-PTC-REQ-054` – **Analysis/gridding coefficient.** Each analysis/gridding family shall declare detector or detector-time index family, exact statistic/factors, their domains, unit, normalization operator, normalization domain, selected support, lifecycle, permitted consumers, numerical use, and prohibited interpretations.

`SCI-PTC-REQ-055` – **Coefficient re-estimation.** Re-estimating any coefficient shall create a new realized product state and shall not silently modify an earlier transformed signal or parent.

`SCI-PTC-REQ-056` – **Empirical statistic limit.** An empirical residual statistic shall declare centering, coefficients, support, effective degrees of freedom, correlation treatment, and lifecycle and shall not be labeled unbiased variance or precision without proof.

`SCI-PTC-REQ-057` – **Uncertainty taxonomy.** Formal covariance, marginal variance, empirical scatter, calibration/systematic uncertainty, response uncertainty, selection uncertainty, and cross-coordinate covariance shall be distinct typed states.

`SCI-PTC-REQ-058` – **Conditional covariance.** A covariance propagated through $J_\Theta[Y]$, the derivative or linear part of the exact frozen application map including frozen λ, shall be labeled conditional on the exact resolved selected model and shall identify axes, units, normalization, support, rank, approximation, and omitted correlations.

`SCI-PTC-REQ-059` – **Selection uncertainty.** When fitted modes, grouping, rank, masks, classification, or coefficients vary, omission of the between-selection term in Equation 21 shall be declared; unknown selection uncertainty shall not be zero.

`SCI-PTC-REQ-060` – **Cross-observation covariance.** An absent cross-observation covariance block shall mean unknown/unavailable, not independence; shared calibration, atmosphere, model, APT, or processing state shall remain possible correlation causes.

9.7 Response and transfer

`SCI-PTC-REQ-061` – **Admitted upstream response.** A complete chain response shall begin from $K_{\mathrm{up}\to\mathrm{CAL}}$, the complete admitted upstream response carried by the CAL product, with explicit source domain and CAL detector-time codomain. It shall compose every applicable admitted source-to-detector relation, beam convention, coordinate and scan geometry, RTC operation, and CAL operation, or the affected complete-response claim shall be unavailable. A supplied companion already on the CAL grid shall instead enter the PTC-local operator directly.

`SCI-PTC-REQ-062` – **Fixed-state companion.** A fixed-state companion shall be acted on by $J_{\Theta_g}[Y_g]$, the derivative or linear part of the exact frozen group-local application map, with the same group, coordinates, metric, supports, masks, strictly positive rank, subspace, generalized inverse/tolerance, detector classes, and boundaries. It shall not enter or alter learning; frozen λ_g shall not be subtracted from the perturbation, re-estimated, or restored; and $K_{\mathrm{up}\to\mathrm{CAL}}$ shall not be applied to a CAL-grid companion a second time.

`SCI-PTC-REQ-063` – **Full-procedure response.** A full-procedure response shall rerun complete PTC centering, fitting, selection, rank, classification, refinement, and application from the immutable admitted CAL parent under a declared perturbation. The location λ shall be re-estimated for each run, retained as changed state, and discarded from each conditioned output. Its numerical finite difference shall use only the transformed-signal projection $\mathcal{P}_Z$; changed discrete or mixed state shall be retained separately through Δ_{state}.

`SCI-PTC-REQ-064` – **Response family.** Where response depends materially on amplitude, perturbation side, location, source state, mask, rank, classification, or convergence, PTC shall publish a bounded response family or mark a stronger response unavailable.

`SCI-PTC-REQ-065` – **Package response boundary.** A whole-chain RTC-to-CAL-to-PTC injection shall be a separately named cross-package study and shall not be relabeled PTC-only full-procedure response.

`SCI-PTC-REQ-066` – **Response status.** Every requested response role shall use exactly one availability status among computed/published, not computed or not requested for this product, invalid, and unavailable, with cause and validity domain. Response availability shall remain distinct from product realization state.

`SCI-PTC-REQ-067` – **Map-center diagnostic.** The optional estimated map-center point-source response shall require an exact source companion and parent domain, explicit fixed-state or full-procedure response role, complete propagated PTC companion, exact named reference MAP operator, center selector, band/mode/state binding, and uncertainty status.

`SCI-PTC-REQ-068` – **No response overclaim.** A map-center or compact-source response shall not establish general PTC, off-center, spatially varying, extended-source, arbitrary-morphology, cross-band, cross-mode, beam-shape, or MAP authority.

9.8 Outputs, provenance, and boundaries

`SCI-PTC-REQ-069` – **Transformed intermediate.** The in-memory transformed TOD shall be the authoritative PTC-to-MAP intermediate on the PTC-dependent route and shall not be labeled an independent sky estimator.

`SCI-PTC-REQ-070` – **Persisted role.** A persisted PTC TOD shall declare either diagnostic artifact or requested derived analysis product; persistence shall not grant archival, map, beam, significance, or production authority.

`SCI-PTC-REQ-071` – **Material provenance.** All material state affecting a declared consumer shall be linked to immutable parent, requested, effective-policy, observation-resolved, learned/fitted evidence, resolved selected model, applied/realized, and published states, plus supports, groups, modes, rank, centering location and its nonrestoration, coefficients, classification, response, covariance, randomness, and output identity.

`SCI-PTC-REQ-072` – **Atomic required outputs.** A required product shall be complete only after all required components and links are durable. Partial publication or required-output failure shall propagate and shall not be labeled complete.

`SCI-PTC-REQ-073` – **Typed fallback.** Every separately authorized fallback shall declare cause, stage, changed estimator/state, response effect, covariance effect, and affected availability; unavailable support shall not become valid zero output. The ordinary route permits no rank, support, centering-only, or route-conversion fallback.

`SCI-PTC-REQ-074` – **Insufficient support.** Empty, non-finite, zero-denominator, unresolved-degeneracy, or otherwise insufficient group-local support shall fail the ordinary requested rank with exact cause and no clipping or substitution. Another family may use a fallback only when explicitly authorized and typed.

`SCI-PTC-REQ-075` – **Random state.** Randomness capable of changing science output shall record generator family/version, seed, stream partition, draw identity, and input identity; repeated declared state shall be reproducible.

`SCI-PTC-REQ-076` – **RTC-terminal disabled route.** An explicit PTC-disabled request shall select the RTC-terminal export route. Citlali shall complete and publish the required consumer-neutral RTC bundle, then terminate successfully without entering CAL, PTC, or MAP and without fabricating products for those stages. Failure of required RTC publication shall fail the route.

`SCI-PTC-REQ-077` – **No ordinary PTC-free map route.** Ordinary Citlali map production shall require a realized PTC transformed timestream. PTC-disabled and invalid rank shall produce no map and shall not select a direct CAL-to-MAP fallback. This route exclusion supplies no MAP estimator, projection, coefficient, support, or product authority.

`SCI-PTC-REQ-078` – **Excluded signal roles.** Base v0.1 shall not authorize raw-$\Delta f/f$ Beammap PTC, raw-r numerical science, r-derived subtraction from x, unconstrained joint x/r PCA, cross-array fitting, or polarimetry.

`SCI-PTC-REQ-079` – **Ownership preservation.** Consuming another package's product shall not transfer ownership of RTC temporal conditioning, CAL calibration/atmosphere, AST/ALIGN coordinates, shared VAL types/evaluation machinery, any named-use owner's admission policy, MAP estimation, NOI uncertainty, BEAM interpretation, FLT filtering, or FRUIT recurrence.

`SCI-PTC-REQ-080` – **Evidence-layer separation.** The contract shall distinguish algebraic contract correctness, implementation conformity, representation/response fidelity, observational performance, and production readiness. This document shall claim only the first.

9.9 Explicit scientific-review scope-completion obligations

`SCI-PTC-REQ-081` – **External source/residual parent.** Any source model, mask, residual, or prior-pass parent consumed for protection or fitting shall declare owner, product and recurrence identity, unit, coordinate frame, registration, validity, response, use-specific support, boundary rule, and relation to the immutable CAL parent. Missing required identity shall make the affected protected or fitted role unavailable.

`SCI-PTC-REQ-082` – **Shifted/null surrogate.** A shifted or null surrogate shall transform signal and its associated producer facts, causes, support decisions, boundaries, randomness, and parent identity together under one declared rule. Insufficient transformed support shall be unavailable or rejected and shall not become a valid numerical-zero surrogate.

`SCI-PTC-REQ-083` – **Exact frozen application map.** Every realized estimator shall publish $\mathcal{A}_{\Theta_g}$ and $J_{\Theta_g}[Y_g]$ with exact group-local input/output domains, detector-right action for the ordinary route, frozen quantities including λ_g, mask-aware recomputed coefficients, metric, supports, rank, subspace, generalized inverse/tolerance, affine reference, and failure behavior. The map shall subtract and not restore λ_g. It shall define $U_{\mathrm{total},\Theta} = Y - \mathcal{A}_\Theta(Y)$; equality to subtraction of $\widehat{U}_{\mathrm{corr}}$ alone shall not substitute for this contract.

`SCI-PTC-REQ-084` – **Candidate evidence retention.** For the ordinary route, learned/fitted evidence shall retain the one explicitly requested positive-rank candidate-model specification, group-local fitted or failed state, feasibility cause, diagnostics, and exact immutable application map. A future automatic-selection family shall additionally retain every evaluated candidate, predicate value and pass/fail cause, ordering and tie result, and selected or rejected disposition.

`SCI-PTC-REQ-085` – **Typed astronomical-transfer predicate.** An astronomical-transfer predicate is not requested for ordinary configured-rank admission. Any future requested predicate shall identify source class and companion, conditional or full-procedure response type, amplitude and location domain, metric, tolerance, mask/support, upstream response identity, response-uncertainty treatment, and exact claim tested.

`SCI-PTC-REQ-086` – **Expectation and bias statement.** Each estimator family and product role shall state its conditional expectation or bias claim for astronomical signal, fitted nuisance, remaining detector noise, calibration uncertainty, and data-dependent selection, or mark each unsupported claim unavailable. No estimator shall be labeled unbiased by omission.

`SCI-PTC-REQ-087` – **Response-domain chain.** Every response product shall distinguish the PTC-local operator $J_\Theta[Y]$, complete chain response $J_\Theta[Y] \circ K_{\mathrm{up}\to\mathrm{CAL}}$, and realized propagated companion. Source-domain and CAL-grid parents shall have explicit domains, the response carried by the CAL product shall not be confused with only the CAL-owned multiplier, and each upstream operator shall be applied exactly once.

`SCI-PTC-REQ-088` – **Independent state axes.** For a producing stage that is entered, product realization, response availability, covariance availability, and validation/evidence disposition shall be separate typed axes. The RTC-terminal export route does not enter PTC and shall not fabricate PTC product, response, or covariance states. For a realized PTC product, unavailable response shall not imply that its otherwise supported signal does not exist.

`SCI-PTC-REQ-089` – **Fit-excluded application availability.** A fit-excluded occurrence shall be application-available only when the resolved group-local frozen operator defines its value from admitted fitted state and support and every required loading, time-local coefficient-recomputation input, application mask, detector binding, metric term, coordinate transform, and boundary state is available within the same configured network or array group. Otherwise application shall be unavailable; it shall not be interpolated, zero-filled, reconstructed, or borrowed from another group.

9.10 Ordinary configured-rank route completion

`SCI-PTC-REQ-090` – **Ordinary route identity.** The first ordinary route shall be one explicitly requested calibrated-x, configured-rank, group-local PCA/SVD operation with exact CAL parent, segment, output identity, lifecycle, and failure state.

`SCI-PTC-REQ-091` – **Mutually exclusive grouping level.** Each ordinary route instance shall select array-level or network-level PCA, never both by implication. Array mode shall realize one group spanning the declared array; network mode shall realize one independent group per declared readout network within that array.

`SCI-PTC-REQ-092` – **Group-local scientific state.** Every fit/loading/application/output support, mask, centering state, fitted subspace, rank, coefficient solve, transformed output, kernel response, failure, and provenance item shall carry the exact configured group identity. A network-level support change shall not alter another network's resolved operator.

`SCI-PTC-REQ-093` – **Ordinary centering and scaling.** Within each exact immutable PTC segment and configured group, the ordinary route shall compute $\lambda_{g,d}$ as the support-normalized arithmetic mean of the finite basis-fit-admitted samples for each detector, using binary centering influence, subtract and publish that location without restoration, and use identity internal scaling. No sample across a segment boundary, gap, network, or array shall enter the estimator. A zero, nonfinite, or otherwise invalid denominator or insufficient centering support shall fail the affected group without an inferred estimator, fallback, or borrowed support.

`SCI-PTC-REQ-094` – **Explicit positive rank.** The ordinary request shall supply an integer rank $k_{\mathrm{req},g} \geq 1$ for each group or one explicitly common positive rank applied to named groups. Each group shall independently verify that the requested rank is realizable on its exact support and degeneracy state.

`SCI-PTC-REQ-095` – **Rank failure is not routing.** Rank zero, noninteger rank, or unrealizable positive rank shall fail closed with exact group and cause. It shall not yield centering-only output, be clipped or substituted, become PTC-disabled, select RTC-terminal export, or produce a map.

`SCI-PTC-REQ-096` – **One immutable-parent fit.** The ordinary route shall perform exactly one fit per configured group from the immutable admitted CAL parent and zero support-changing refinements. Advisory diagnostics shall not modify the fitted evidence, support, rank, resolved model, or application state.

`SCI-PTC-REQ-097` – **Group-local fixed kernel.** A requested compatible CAL-grid kernel shall be acted on by the same group-local fixed-state linear operator, application mask, metric, strictly positive rank, subspace, generalized inverse/tolerance, and support as the data. It shall not enter learning, alter the science result, or have the frozen λ_g subtracted from it.

`SCI-PTC-REQ-098` – **Classification preservation.** PTC shall preserve the exact upstream CAL classification, including `engineering-only`. Classification alone shall not prohibit PTC mathematics; the owner of each exact use shall decide admission. PTC shall not relabel engineering-only input into CAL science-qualification eligibility, and no separate engineering route is required unless a later differing policy requires one.

`SCI-PTC-REQ-099` – **Ordinary route exclusions.** The first ordinary route shall perform no automatic rank selection, source protection, support-changing refit, conditioned-r influence, simulation/empirical population, FRUIT recurrence, full-procedure response, or stronger covariance claim. A local fixed-state kernel may be available without establishing any stronger response or uncertainty tier.

10 Falsifiable edge predictions

These are scientific oracles for later conformance and validation work. They are not claims that an implementation has passed.

`SCI-PTC-PRED-001` – **RTC-terminal disabled route.** With PTC explicitly disabled, complete required RTC publication is followed by successful Citlali termination; CAL, PTC, and MAP are not entered and no product states for those stages are fabricated. Failure of required RTC publication fails the route.

`SCI-PTC-PRED-002` – **Same-unit response loss.** A detector-common astronomical perturbation can retain mJy-per-fixed-nominal-beam units while being attenuated or annihilated; unit equality alone will not predict recovered amplitude.

`SCI-PTC-PRED-003` – **Correlation non-identification.** Two data sets with the same detector covariance but different physical mixtures admit the same fitted correlated subspace; the fit alone cannot distinguish atmosphere from sky or electronics.

`SCI-PTC-PRED-004` – **Gauge invariance.** Negating a basis/loading pair, reciprocally rescaling it, or rotating a degenerate basis leaves the fitted correlated signal and transformed output unchanged within declared numerical tolerance.

`SCI-PTC-PRED-005` – **Centering null mode.** On unchanged finite centering support, adding a constant perturbation to every admitted occurrence of one detector shifts its support-normalized arithmetic mean by the same constant and leaves that detector's centered fit coordinate unchanged in a full-procedure rerun; the re-estimated location is again discarded. In fixed-state response λ is held fixed and the perturbation is propagated by the already-fitted $J_\Theta[Y]$.

`SCI-PTC-PRED-006` – **Scaling restoration.** For an invertible internal scaling multiplied by a finite nonzero constant and consistently inverted, ordinary transformed output in the physical unit is unchanged within tolerance.

`SCI-PTC-PRED-007` – **Zero-fill is not exclusion.** Replacing one excluded nonzero sample by zero before ordinary PCA changes means/covariance and can change the selected subspace; a correctly masked/weighted missing-data estimator gives zero fit influence instead.

`SCI-PTC-PRED-008` – **Fit-excluded apply-allowed.** An occurrence excluded from fitting but admitted for application is acted on by the exact frozen application map without affecting learned or selected state; toggling only its output support does not refit.
`SCI-PTC-PRED-009` – **Support-stage independence.** Changing coefficient-only support changes the affected coefficient product but not the fitted basis or transformed signal; changing fit support invalidates or refits the fitted state.
`SCI-PTC-PRED-010` – **Influence through fitting.** A single occurrence used to fit a common mode can influence subtractions at many other occurrences even where its direct local derivative is zero; those outputs carry transitive fit influence.
`SCI-PTC-PRED-011` – **Source-mask boundary.** Moving a sample across a declared mask boundary changes fit/application behavior only according to the serialized polarity and boundary rule; unmasked sky remains eligible to project into the removed subspace.
`SCI-PTC-PRED-012` – **Configured-group isolation.** Changing data in another array does not change the selected array's result. In network mode, changing one network does not change another network's fitted support, subspace, or kernel operator; in array mode the same change may alter the one array-wide subspace.
`SCI-PTC-PRED-013` – **Grouping alternatives.** The same array can yield different removed subspaces under configured network- and array-level PCA because their support and coupling domains differ. Neither result is reproduced by silently applying both levels sequentially.
`SCI-PTC-PRED-014` – **Common-mode denominator failure.** A group/time with zero, empty, or non-finite admitted denominator yields unavailable/rejected state or a typed fallback, never a silently valid zero common mode.
`SCI-PTC-PRED-015` – **Fixed-template orthogonality.** For a frozen fixed-template metric, a perturbation exactly orthogonal to the removed subspace is preserved by the conditional operator; learned-template motion is a separate full-procedure effect.
`SCI-PTC-PRED-016` – **Rank-deficient gauge.** Duplicate or zero columns produce a rank-deficient fit whose generalized inverse, tolerance, selected subspace, and deterministic tie rule reproduce the emitted subtraction.
`SCI-PTC-PRED-017` – **Future least-aggressive selection.** For a separately authorized automatic-selection family with nested candidates where the first two fail at least one required predicate and the third passes all, the third is selected even if an earlier candidate has a better scalar score. The ordinary configured-rank route does not perform this comparison.
`SCI-PTC-PRED-018` – **Future no-compensating metric.** In a separately authorized automatic-selection family, excellent residual suppression cannot compensate for a failed astronomical-transfer predicate. No diagnostic changes the ordinary route's explicitly requested rank.
`SCI-PTC-PRED-019` – **Future deterministic tie.** Two future automatic-selection candidates equal under every required predicate and ordering resolve identically using the registered tie rule. The singleton ordinary candidate has no data-derived tie decision.
`SCI-PTC-PRED-020` – **Failed candidate resolution.** An unrealizable ordinary requested rank produces no transformed signal or map and is not clipped. If no candidate passes a future automatic-selection family, no transformed product claiming that admission is produced.
`SCI-PTC-PRED-021` – **Immutable-parent refinement.** The ordinary route emits one immutable-parent fit and no second support-changing fit. In a separately authorized future refinement, a second fit equals a fresh fit of the complete selected model on the immutable CAL parent and differs, in general, from fitting the already cleaned first output.
`SCI-PTC-PRED-022` – **Output-only classification.** Changing one detector from retained to postfit-output-reject without changing fit support preserves the fitted subspace and other detectors' transformed values.
`SCI-PTC-PRED-023` – **Refinement oscillation.** The ordinary route cannot oscillate through support-changing refits because its maximum is zero. A future refinement with alternating high-leverage membership reaches its declared oscillation/exhaustion state and is never labeled converged.
`SCI-PTC-PRED-024` – **Diagnostic population dependence.** The same residual statistic can differ under two declared reference populations, but in the ordinary route neither diagnostic result changes support, configured rank, or resolved model.
`SCI-PTC-PRED-025` – **Diagnostic-only r inertia.** Changing an admitted conditioned-r diagnostic while holding calibrated x and all x policy fixed can change only r diagnostic outputs, not x fit membership, configured or resolved rank, subtraction, output, or coefficients.
`SCI-PTC-PRED-026` – **Raw-r incompatibility.** If conditioned-r operator, unit, response, optical-leakage state, or exact x-grid relation is missing, the r diagnostic branch is unavailable while the admitted x branch may remain defined.

`SCI-PTC-PRED-027` – **Coefficient gauge.** Reciprocal basis/loading rescaling changes fitted loading values but leaves the modeled subtraction unchanged; an analysis/gridding coefficient does not transform merely because a loading gauge changes.
`SCI-PTC-PRED-028` – **Coefficient normalization.** Multiplying all raw gridding coefficients by a common nonzero factor can leave a normalized weighted estimate unchanged while changing the denominator, proving that the denominator is not automatically precision.
`SCI-PTC-PRED-029` – **Conditional covariance.** For a fixed analytic $J_\Theta[Y]$ and declared C_Y, the conditional covariance equals $J_\Theta[Y]C_Y J_\Theta[Y]^\mathsf{T}$ including off-diagonal detector/time terms.
`SCI-PTC-PRED-030` – **Selection variance term.** Across perturbations that change rank or support, total output variance exceeds or differs from the mean conditional covariance whenever between-selection means vary.
`SCI-PTC-PRED-031` – **Fixed companion noninterference.** Adding a response companion through the fixed-state path leaves fitted modes, rank, masks, detector classes, and transformed science data unchanged.
`SCI-PTC-PRED-032` – **Full-procedure nonlinearity.** An injected source that crosses a rank, mask, clipping, classification, or convergence boundary produces amplitude- or side-dependent finite response rather than one global linear kernel.
`SCI-PTC-PRED-033` – **Package response boundary.** Rerunning PTC from a fixed CAL parent can characterize PTC-full-procedure response but cannot expose a changed RTC or CAL learned state; that requires a separately named whole-chain study.
`SCI-PTC-PRED-034` – **Map-center factor availability.** Removing the exact source companion or named reference MAP operator makes the map-center diagnostic unavailable while leaving a supported sample-domain PTC response potentially available.
`SCI-PTC-PRED-035` – **Compact versus extended response.** A compact source and detector-common extended mode with equal nominal peak amplitude can have different PTC responses; the compact map-center diagnostic cannot predict the extended response.
`SCI-PTC-PRED-036` – **Random-state reproducibility.** Repeating a simulation or randomized estimator with identical generator/version, seed, stream, and input identity reproduces the declared draws and realized result; changing one stream changes only its assigned draws.
`SCI-PTC-PRED-037` – **Required-output atomicity.** Failure to publish one required sibling or link leaves the product explicitly incomplete; a remaining readable file cannot be labeled complete.
`SCI-PTC-PRED-038` – **Evidence-layer separation.** Successful algebraic and document-coverage checks do not change implementation conformity, representation fidelity, observational performance, or production-readiness status from unassessed/not performed.
`SCI-PTC-PRED-039` – **Frozen component versus projection.** For a perturbation with nonzero projection onto the frozen removed subspace, subtraction of the typed frozen total-removed operand has identity derivative while frozen-subspace application removes that projected component; the two responses differ even though both reproduce the fitted-parent transformed result.
`SCI-PTC-PRED-040` – **Temporal versus detector action.** For a nonsymmetric response matrix H, temporal-left projection $\mathcal{L}^t_\Theta H$ and detector-right projection $H\mathcal{L}^d_\Theta$ generally differ; matrix orientation and rank alone cannot select the propagated companion.
`SCI-PTC-PRED-041` – **No double upstream response.** Propagating a CAL-grid companion through $J_\Theta[Y]$ gives the same result as propagating its source-domain parent through $J_\Theta[Y] \circ K_{\mathrm{up}\to\mathrm{CAL}}$ exactly once; applying $K_{\mathrm{up}\to\mathrm{CAL}}$ again changes the domain or value and is rejected.
`SCI-PTC-PRED-042` – **Typed procedure state change.** A perturbation that changes selected rank or learned centering location yields a valid transformed-signal finite difference and separate typed state changes; attempting to subtract the complete procedure tuples is undefined and cannot produce the response.
`SCI-PTC-PRED-043` – **Candidate-evidence reconstruction.** Given the retained candidate order, fitted states, predicate values, causes, and tie rule, the selected or rejected disposition is reproduced exactly; removing one failed predicate record makes the selection audit incomplete.
`SCI-PTC-PRED-044` – **External-parent fail-closed.** Removing the recurrence identity or coordinate registration from an otherwise numerical source/residual parent makes the affected protected or fitted role unavailable while leaving independent PTC roles unchanged.
`SCI-PTC-PRED-045` – **Surrogate support transformation.** Shifting surrogate signal without shifting its associated facts, causes, support decisions, boundaries, and identity changes the admitted population and fails the surrogate contract; shifting them together reproduces the declared transformed membership, while insufficient support is rejected rather than zero-filled.

`SCI-PTC-PRED-046` – **Astronomical-predicate typing.** Two candidates with the same scalar transfer value but different source class, response type, amplitude/location domain, or uncertainty treatment do not satisfy the same astronomical-transfer predicate and cannot be substituted.
`SCI-PTC-PRED-047` – **Unavailable bias claim.** When an estimator family supplies no expectation/bias statement for data-dependent selection, the bias claim remains unavailable even if one synthetic realization has zero measured residual mean.
`SCI-PTC-PRED-048` – **Independent product and response states.** A realized transformed signal can coexist with unavailable complete response. The RTC-terminal route never enters PTC and therefore has no PTC response axis to encode; neither condition can be inferred from the other.
`SCI-PTC-PRED-049` – **Undefined fit-excluded extension.** For a fit-excluded occurrence missing any basis coefficient or coefficient-recomputation input, neighboring support, detector binding, metric term, coordinate transform, or boundary state required by the resolved operator, application is unavailable. Filling, interpolating, or reconstructing that occurrence changes the declared operator and cannot reproduce the authorized fit-excluded/apply-allowed result.
`SCI-PTC-PRED-050` – **Conservative support composition.** For an exact PTC-owned use, adding an applicable excluding cause to otherwise permitting facts changes the composite result to ineligible and cannot be rescued by another permitting cause. With no known exclusion but one unknown required predicate, the result is `decision_unavailable`; supplying that predicate as true can then make the result eligible without erasing any cause.
`SCI-PTC-PRED-051` – **Network support isolation.** In network mode, changing fit or application support in one network changes only that network's fitted or applied state; every other network's Θ_g and propagated kernel remain identical.
`SCI-PTC-PRED-052` – **Array cross-network coupling.** In array mode, changing admitted data in one network can change the one array-wide loading subspace and therefore the transformed data or kernel in another network within the same array.
`SCI-PTC-PRED-053` – **Rank-zero failure.** A request with $k = 0$ yields an explicit failed PTC resolution, no centering-only output, no RTC-export route conversion, and no map.
`SCI-PTC-PRED-054` – **Unrealizable-rank failure.** A positive rank exceeding exact group-local feasibility yields failure with group and cause; the emitted state contains no clipped or substituted rank and no transformed output.
`SCI-PTC-PRED-055` – **Configured-rank reproducibility.** Repeated resolution of the same CAL parent, configured group, support, masks, metric, degeneracy tolerance, and positive requested rank reproduces the same removed subspace and operator within declared numerical tolerance.
`SCI-PTC-PRED-056` – **Full-support projector reduction.** When every detector in a group has uniform application influence, the mask-aware coefficient solve reduces within tolerance to detector-right projection through $I - \widehat{A}_g(\widehat{A}_g^{\top}\widehat{A}_g)^{+}\widehat{A}_g^{\top}$.
`SCI-PTC-PRED-057` – **Masked influence is not zero data.** Changing the numerical value stored at an application-excluded detector while keeping its influence zero leaves the time-local coefficient solve unchanged; changing its influence to admitted can change the solve. Replacing the excluded value by an admitted numerical zero need not reproduce either result.
`SCI-PTC-PRED-058` – **Data-kernel state identity.** For fixed Θ_g, applying the group-local linear operator to a compatible kernel uses the same support, mask, metric, rank, subspace, generalized inverse, and tolerance as the data, leaves all learned state unchanged, and does not subtract λ_g from the perturbation.
`SCI-PTC-PRED-059` – **Advisory-diagnostic inertia.** Changing an ordinary post-fit advisory diagnostic while holding the immutable CAL parent and resolved operator state fixed changes only that diagnostic record; support, rank, Θ_g, transformed data, and propagated kernel remain unchanged.